

EEZ Aquaculture Development

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Prioritizing Potential Aquaculture

This repository investigates temperature and depth requirements for different aquaculture species on the West Coast.

Loading Libraries

```
library(tidyverse)
library(terra)
library(stars)
library(sf)
library(tmap)
library(kableExtra)
```

Downloading data - Temperature, depth, and West Coast EEZ data

```
# Sea Surface Temperature
rast_2008 <- rast(here::here('data', 'average_annual_sst_2008.tif'))
rast_2009 <- rast(here::here('data', 'average_annual_sst_2009.tif'))
rast_2010 <- rast(here::here('data', 'average_annual_sst_2010.tif'))
rast_2011 <- rast(here::here('data', 'average_annual_sst_2011.tif'))
rast_2012 <- rast(here::here('data', 'average_annual_sst_2012.tif'))

# Depth
bathy <- rast(here::here('data', 'depth.tif')) %>%
  project('EPSG:4326')

# Exclusive Economic Zones
eez <- read_sf(here::here('data', 'wc_regions_clean.shp'))
```

Coordinate Reference System

Check if the CRS match and update if not

```
# CRS UPDATING function
crs_update <- function (df1, df2) {
  if (st_crs(df1) == st_crs(df2)) {
    print("CRS match")
  } else {
    warning("Updating CRS")
    df1 <- st_transform(df1, crs = "EPSG:4326")
    df2 <- st_transform(df2, crs = "EPSG:4326")
  }
}

# CRS CHECK function (with warning)
crs_check <- function (df1, df2) {
  if (st_crs(df1) == st_crs(df2)) {
    print("CRS match")
  } else {
    warning("CRS do NOT match")
  }
}

# Check the dataframes
crs_update(bathy, eez)
```

```
[1] "CRS match"
```

```
crs_update(bathy, rast_2008)
```

```
[1] "CRS match"
```

```
crs_update(rast_2008, rast_2009)
```

```
[1] "CRS match"
```

```
crs_update(rast_2009, rast_2010)
```

```
[1] "CRS match"
```

```
crs_update(rast_2010, rast_2011)
```

```
[1] "CRS match"
```

```
crs_update(rast_2011, rast_2012)
```

```
[1] "CRS match"
```

Process data

Combine sst rasters

```
# Combine Sea Surface Temperature Rasters
sst_rast <- c(rast_2008, rast_2009, rast_2010, rast_2011, rast_2012)
# rast() would also work to merge

# Remove unbinned rasters
rm(list = c('rast_2008', 'rast_2009', 'rast_2010', 'rast_2011', 'rast_2012'))
```

Find the **mean** SST - local algebra

```
# Find the mean SST
sst_mean <- mean(sst_rast)
```

Convert SST from Kelvin to Celsius - global algebra

```
# Subtract 273.15
sst_mean <- sst_mean - 273.15
```

Crop depth raster to match the *extent* of the SST raster

```
# Extent of sst raster
sst_ext <- ext(sst_mean)

# Crop bathy to sst raster extent
bathy_crop <- crop(bathy, sst_ext)

rm(sst_ext)
```

Fixing unmatching resolutions

```
# Resample bathy / depth data to match sst raster
bathy_updated <- resample(bathy_crop, sst_mean, method = "bilinear")

# Check resolutions
if (all(res(sst_rast) == res(bathy_updated))) {
  print("Resolutions match")
} else {
  warning("Resolutions don't match!")
}
```

```
[1] "Resolutions match"
```

```
# Remove uncropped and unresolutioned depth dfs
rm(list = c('bathy', 'bathy_crop'))
```

Final Check depth and SST dfs match in resolution, extent, and coordinate reference system

```
# Do resolutions match?
all(res(bathy_updated) == res(sst_mean)) # Equal in both vs all directions
```

```
[1] TRUE
```

```
# Do extents match?
ext(bathy_updated) == ext(sst_mean)
```

```
[1] TRUE
```

```
# Do CRS match?
crs_check(bathy_updated, sst_mean)
```

```
[1] "CRS match"
```

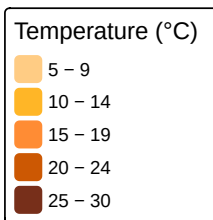
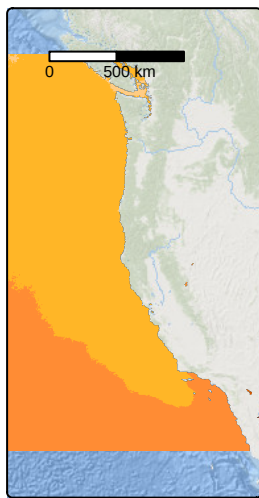
Temp and Depth Map of West Coast

```
# Map sst and bathy
bath_map <- tm_shape(bathy_updated) +
  tm_raster(col.legend = tm_legend(title = "Depth (m)")) +
  tm_title(text = "Depth (m) of West Coast") +
  tm_compass(position = c('right', 'top'))

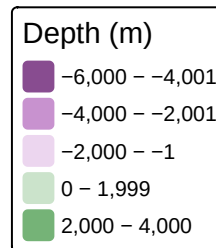
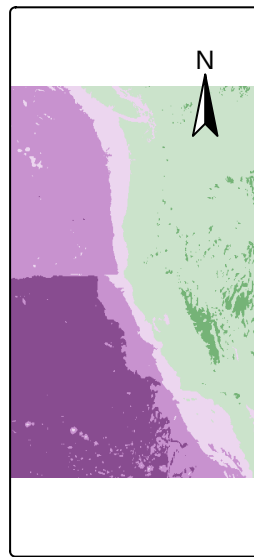
sst_map <- tm_shape(sst_mean) +
  tm_raster(col.legend = tm_legend(title = "Temperature (°C)",
    col.scale = tm_scale(values = c('#ffcc84', "#FFB627", "#ff8c34", "#CC5803", "#772196"))),
  tm_title(text = "Average Sea Surface\nTemperature of West Coast") +
  tm_scalebar(position = c('right', 'top')) +
  tm_basemap("Esri.OceanBasemap")

tmap_arrange(sst_map, bath_map)
```

Average Sea Surface
Temperature of West Coast



Depth (m) of West Coast



Find suitable locations for Oysters

Oysters Research has shown that oysters need the following conditions for optimal growth:

- Sea surface temperature: 11-30°C
- Depth: 0-70 meters below sea level

Reclassifying locations that have optimal oyster temp or depth

- Unsuitable habitats = 0
- Suitable habitats = 1

Suitable **DEPTH** (0-70 meters below sea level) vs Unsuitable locations for oysters

```
# Create a reclassification matrix of values corresponding to unsuitable (1) vs suitable (2)
rcl_depth <- matrix(c(-Inf, -70, 0, # SUITABLE depth
                    -70, 0, 1, # SUITABLE depth
                    0, Inf, 0), # SUITABLE depth
                  ncol = 3, byrow = TRUE)

# Apply the reclassification matrix to the raster
depth_reclass <- classify(bathy_updated, rcl_depth) #other = NA)

# Delete matrix as object
rm(rcl_depth)
```

Suitable **TEMP** (11-30°C) vs Unsuitable locations for oysters

```
# Create a reclassification matrix of values corresponding to unsuitable (1) vs suitable (2)
rcl_temp <- matrix(c(-Inf, 11, 0, # UNSUITABLE temp
                   11, 30, 1, #SUITABLE temp
                   30, Inf, 0), #UNSUITABLE temp
                  ncol = 3, byrow = TRUE)

# Apply the reclassification matrix to the raster
temp_reclass <- classify(sst_mean, rcl_temp) #other =NA)

# Delete matrix as object
rm(rcl_temp)
```

Finding which areas have suitable depth *AND* temp for Oysters

Multiplying temp and depth raster together

```
# Check CRSs
crs_check(temp_reclass, depth_reclass)
```

```
[1] "CRS match"
```

```
# Multiply rasters together
oyst_suit_raster <- temp_reclass * depth_reclass

# OTHER METHOD: FUNCTION WITH LAPP
# multiply <- function(layer1, layer2) {
#   layer1 * layer2
# }
# oyst_suit_raster <- lapp(c(temp_reclass,depth_reclass), fun = multiply)
```

Visualizing depth and temp suitability for oysters in the west coast

```
# Plotting suitable habitats for Oysters
map_temp <- tm_shape(temp_reclass) +
  tm_raster(
    col.scale = tm_scale_categorical(
      labels = c("Unsuitable (0)", "Suitable (1) [11-30°C]"),
      values = c('lightblue2', 'steelblue4'),
    ),
    col.legend = tm_legend(title = "Suitable vs Unsuitable\nOysters Temp")) +
  tm_title(text = "Suitable Temp for Oysters") +
  tm_compass(position = c('right', 'top'))

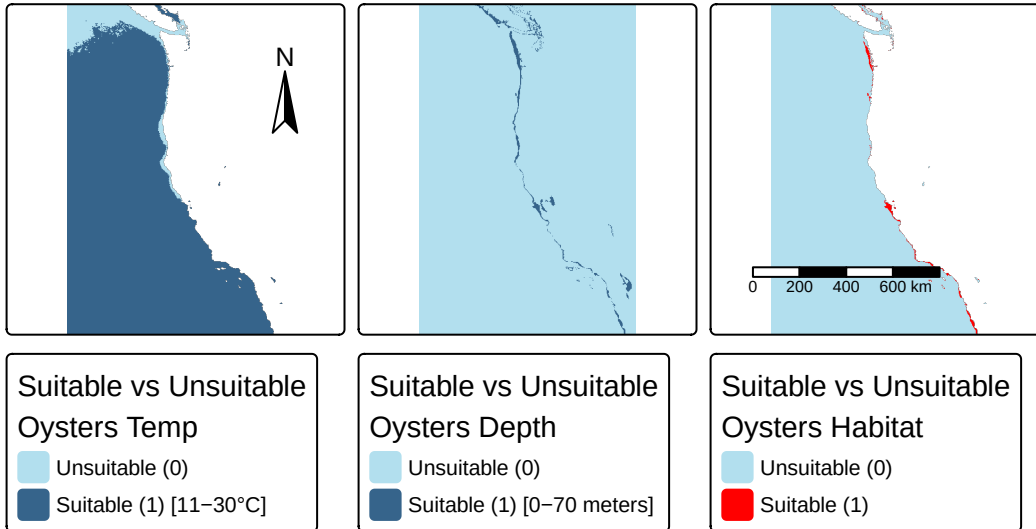
map_depth <- tm_shape(depth_reclass) +
  tm_raster(
    col.scale = tm_scale_categorical(
      labels = c("Unsuitable (0)", "Suitable (1) [0-70 meters]"),
      values = c('lightblue2', 'steelblue4'),
    ),
    col.legend = tm_legend(title = "Suitable vs Unsuitable\nOysters Depth")
  ) +
  tm_title(text = "Suitable Depth for Oysters")

map_oyst <- tm_shape(oyst_suit_raster) +
  tm_raster(
    col.scale = tm_scale_categorical(
      labels = c("Unsuitable (0)", "Suitable (1)"),
      values = c('lightblue2', 'red'),
    ),
    col.legend = tm_legend(title = "Suitable vs Unsuitable\nOysters Habitat")) +
  tm_title(text = "Suitable Depth & Temp for Oysters") +
```

```
tm_scalebar(position = c('left', 'bottom'))

tmap_arrange(map_temp, map_depth, map_oyst)
```

Suitable Temp for Oysters Suitable Depth for Oysters Suitable Depth & Temp for Oysters



```
# Remove maps as objects
rm(list = c('map_temp', 'map_depth', 'map_oyst'))
```

Determining most suitable EEZ

Mask the Oyster raster to the EEZ geometries

```
# Check is crs match
crs_check(oyst_suit_raster, eez)
```

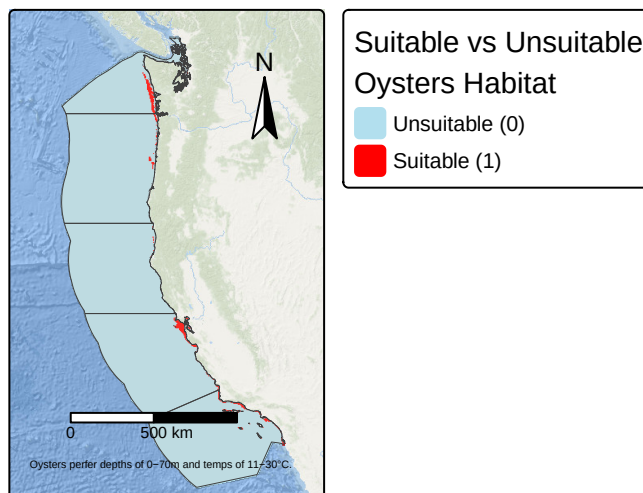
```
[1] "CRS match"
```

```
# Mask suitable hab raster to the EEZ area
oyst_suit_crop <- mask(oyst_suit_raster, eez)
```

Plot Oyster suitability *within* the EEZ areas

```
# Check if it masked
tm_shape(oyst_suit_crop) +
  tm_raster(
    col.scale = tm_scale_categorical(
      labels = c("Unsuitable (0)", "Suitable (1)"),
      values = c('lightblue2', 'red'),
    ),
    col.legend = tm_legend(title = "Suitable vs Unsuitable\nOysters Habitat")) +
  tm_title(text = "Suitable Depth & Temp for Oysters within\nthe West Coast Exclusive Econom.") +
  tm_shape(eez) +
  tm_polygons(
    lwd = 0.25,
    fill = "peachpuff2",
    fill_alpha = 0.2) +
  tm_scalebar(position = c('left', 'bottom'),
    margins = 1
  ) +
  tm_compass(position = c('right', 'top')) +
  tm_credits(text = "Oysters prefer depths of 0-70m and temps of 11-30°C.") +
  tm_basemap("Esri.OceanBasemap")
```

Suitable Depth & Temp for Oysters within
the West Coast Exclusive Economic Zones



Find Area of Suitable habitat in West Coast EEZs

Total Area of all West Coast EEZ

```
# Total area of all West Coast EEZ
print(paste("The total area of all West Coast EEZs is", round(sum(st_area(eez))/1e9,2), "x 10^9 m^2"))
```

```
[1] "The total area of all West Coast EEZs is 820.22 x 10^9 m^2"
```

Total Oyster suitable habitat in West Coast EEZ

```
# Creating raster of JUST the suitable habitats [1]
oyst_1 <- oyst_suit_crop
oyst_1[oyst_1 == 0] <- NA # unsuitable hab are 'NAs'

# Find total area of suitable habitats
area <- expnse(oyst_1)[1,2]
# Expense computes the area covered by raster cells that are not NA

print(paste("The area of all suitable oyster habitats in the West Coast EEZs is", round(area,2), "x 10^9 m^2"))
```

```
[1] "The area of all suitable oyster habitats in the West Coast EEZs is 10.6 x 10^9 m^2"
```

Trying new things

```
for (i in unique(eez$rgn)) {
  zone <- eez %>% filter(rgn == i)
  crs_check(zone, oyst_1)
  oyst_zone <- mask(oyst_1, zone)
  area <- expnse(oyst_zone)[1,2]

  zone[, 'area_suit_hab'] <- NA
  zone$area_suit_hab <- area

  df_list <- list()
  df_list[[i]] <- data.frame(
    rgn = i,
    area_per_eez = paste(round(area / 1e8, 2), "x 10^8")
  )
}
```

```
[1] "CRS match"
```

```
[1] "CRS match"
```

```
[1] "CRS match"
```

```
[1] "CRS match"
```

```
[1] "CRS match"
```

```
eez_plus <- do.call(rbind, df_list)
eez_plus
```

```
      rgn area_per_eez
Washington Washington 25.39 x 10^8
```

```
# table <- eez_plus %>%
#   select(rgn, area_suit_hab) %>%
#   st_drop_geometry()

#table
```

Initial generalized function to find the area of suitable habitat for a species per each West Coast EEZ

```
df_list <- list()

# For loop to get area of each eez
for (i in unique(eez$rgn)) {
  area_eez <- eez %>% filter(rgn == i) #
  spec_per_eez <- mask(oyst_1, area_eez)
  area_per_eez <- exppanse(spec_per_eez)[1, 2]

  # Create dataframe
  df_list[[i]] <- data.frame(
    #rgn = i,
    area = paste(round(area_per_eez / 1e8, 2), "x 10^8")
  )
}
```

```
# The dataframe
df_suit_area_eez <- do.call(rbind, df_list)

# Output a kable table
kable(
  df_suit_area_eez,
  align = 'c',
  col.names = c("EEZ location", "Suitable Habitat Area")
)
```

EEZ location	Suitable Habitat Area
Oregon	11.48 x 10 ⁸
Northern California	1.94 x 10 ⁸
Central California	36.74 x 10 ⁸
Southern California	32.42 x 10 ⁸
Washington	25.39 x 10 ⁸

Oysters have a large amount of suitable habitats in Washington, Central CA, and Southern CA. Central and South California's temperature, ranging from 12°C to 20°C, is fully within the limits of oysters preferences. Additionally, oysters have a depth span of 0-70 meters below sea level, meaning oysters will reside closer to the coastline. The Washington and South California's Exclusive Economic Zones offer a larger continental shelf where oysters can thrive. In conclusion, when mixing oysters depth and temperature preferences, Washington, Central CA, and Southern CA offer the most suitable habitats.

The Function

Create a function that does everything above for you

```
habitat <- function(depth_df,
  temp_df,
  depth_min,
  depth_max,
  temp_min,
  temp_max,
  species_name,
  eez_df) {
```

```

# PART 1: MATRIX, RECLASSIFY, COMBINED, MASK, TOTAL AREA *****
# Reclass matrix - DEPTH
rcl_depth <- matrix(
  c(-Inf, depth_min, 0, # SUITABLE depth
    depth_min, depth_max, 1, # SUITABLE depth
    depth_max, Inf, 0), # SUITABLE depth
  ncol = 3,
  byrow = TRUE
)
# Reclass matrix - TEMP
rcl_temp <- matrix(
  c(-Inf, temp_min, 0, # UNSUITABLE temp
    temp_min, temp_max, 1, #SUITABLE temp
    temp_max, Inf, 0), #UNSUITABLE temp
  ncol = 3,
  byrow = TRUE
)

# Apply the reclassification matrix to the raster
depth_reclass <- classify(depth_df, rcl_depth)
temp_reclass <- classify(temp_df, rcl_temp)

# Check (and change crs if necessary)
if (st_crs(temp_reclass) == st_crs(depth_reclass)) {
  print("CRS match")
} else {
  warning("Updating CRS")
  temp_reclass <- st_transform(temp_reclass, crs = crs(depth_reclass))
}

# Multiply rasters
suit_raster <- temp_reclass * depth_reclass

# Crop to eeZ area
species_raster_pre <- mask(suit_raster, eeZ_df)

# Select only suitable habitats
species_raster <- species_raster_pre
species_raster[species_raster == 0] <- NA

# Area of all suitable habitat

```

```

area <- expanse(species_raster)[1, 2]

print(paste(
  "The area of all suitable",
  species_name,
  "habitats in the West Coast EEZs is",
  round(area / 1e9, 2),
  "x 10^9 m^2"
)
)

# Check (and change crs if necessary)
if (crs(species_raster) == crs(eez_df)) {
  print("CRS match")
} else {
  warning("Updating CRS")
  eez_df <- transform(eez_df, crs = crs(species_raster))
}

# PART 2: FIND AREA PER EEZ --> DATAFRAME *****
# Empty list to feed for loop into after
df_list <- list()

# For loop to get area of each eez
for (i in unique(eez_df$rgn)) {
  area_eez <- eez_df %>% filter(rgn == i) #
  spec_per_eez <- mask(species_raster, area_eez)
  area_per_eez <- expanse(spec_per_eez)[1, 2]

  # Create dataframe
  df_list[[i]] <- data.frame(
    #rgn = i,
    area = area_per_eez
    #paste(round(area_per_eez / 1e8, 2), "x 10^8")
  )
}

# The dataframe for the table
df_suit_area_eez <- do.call(rbind, df_list)
# geodata frame for the map

```

```

eez_df[, 'area_suit_hab'] <- NA # Create new column
eez_df$area_suit_hab <- df_suit_area_eez$area

# Output a kable table
print(kable(
  df_suit_area_eez,
  align = 'c',
  format.args = list(scientific = TRUE, digits = 3),
  col.names = c("EEZ location", "Suitable Habitat Area (m)")
))

# printing geodataframe
#print(eez_df)

# PART 2: MAKING MAPS *****

# DEPTH AND TEMP MAPS

# Plotting suitable habitats for Oysters
map_temp <- tm_shape(temp_reclass) +
  tm_raster(
    col.scale = tm_scale_categorical(
      labels = c("Unsuitable (0)", "Suitable (1)"),
      values = c('lightblue2', 'steelblue4'),
    ),
    col.legend = tm_legend(title = "Suitable vs Unsuitable Temp")
  ) +
  tm_title(text = paste("Suitable Temp for", species_name)) +
  tm_compass(position = c('right', 'top'))

map_depth <- tm_shape(depth_reclass) +
  tm_raster(
    col.scale = tm_scale_categorical(
      labels = c("Unsuitable (0)", "Suitable (1)"),
      values = c('lightblue2', 'steelblue4'),
    ),
    col.legend = tm_legend(title = paste("Suitable vs Unsuitable Depth"))
  )

```

```

) +
tm_title(text = paste("Suitable Depth for", species_name))

map_both <- tm_shape(suit_raster) +
  tm_raster(
    col.scale = tm_scale_categorical(
      labels = c("Unsuitable (0)", "Suitable (1)"),
      values = c('lightblue2', 'red'),
    ),
    col.legend = tm_legend(title = "Suitable vs Unsuitable Habitat")
  ) +
tm_shape(eez_df) +
  tm_polygons(
    fill_alpha = 0,
    col = 'black',
    lwd = 0.2) +
tm_title(text = paste("Suitable Depth & Temp for", species_name)) +
tm_scalebar(position = c('left', 'bottom'))

print(tmap_arrange(map_temp, map_depth, map_both))

# MAIN MAP OF SUITABLE HABITAT
output_map <- tm_shape(eez_df,
  bbox = sf::st_bbox(c(
    xmin = -131, xmax = -110,
    ymin = 30, ymax = 50))) +
tm_polygons(
  fill = 'area_suit_hab',
  fill.legend = tm_legend(title = "Total Suitable\nHabitat Area (m)",
    # orientation = 'landscape',
    # position = tm_pos_out("center", "bottom", pos.h = "center")
  ),
  #fill.scale = tm_scale(values = 'Brewer.yl_gn')
  fill.scale = tm_scale_continuous(n = 4,
    values = 'brewer.YlOrBr'),
) +
tm_title(text = paste("Suitable Habitat for", species_name, "within\nthe West Coast Excl
tm_scalebar(position = c('right', 'top'), margins = 1) +

```

```

tm_compass(position = c('right', 'bottom')) +
tm_basemap("Esri.OceanBasemap") +
tm_text('rgn', size = 0.5) +
tm_credits(text = paste0(species_name," prefer depths of ", depth_min," to ", depth_max,
  tm_layout(legend.outside.position = "bottom")

print(output_map)

}

```

Apply the Function to Oysters - Sea surface temperature: 11-30°C - Depth: 0-70 meters below sea level

```

# Use function
df_suit_area_eez <- habitat(
  depth_df = bathy_updated,
  temp_df = sst_mean,
  depth_min = -70,
  depth_max = 0,
  temp_min = 11,
  temp_max = 30,
  species_name = 'Oysters',
  eeze_df = eeze
)

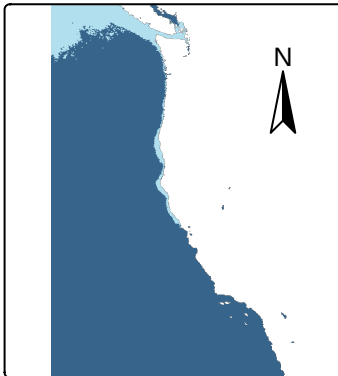
```

```
[1] "CRS match"
```

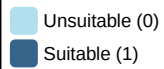
```
[1] "The area of all suitable Oysters habitats in the West Coast EEZs is 10.6 x 10^9 m^2"
```

EEZ location	Suitable Habitat Area (m)
Oregon	1.15e+09
Northern California	1.94e+08
Central California	3.67e+09
Southern California	3.24e+09
Washington	2.54e+09

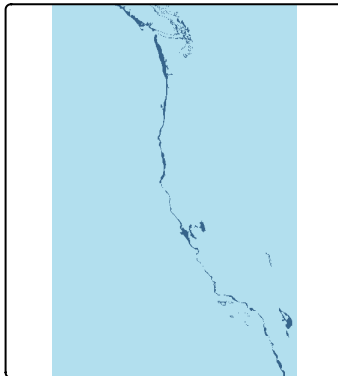
Suitable Temp for Oysters



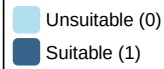
Suitable vs Unsuitable Temp



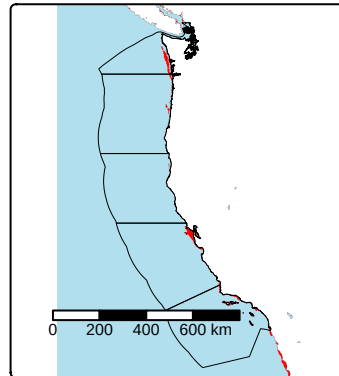
Suitable Depth for Oysters



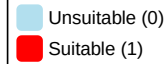
Suitable vs Unsuitable Depth



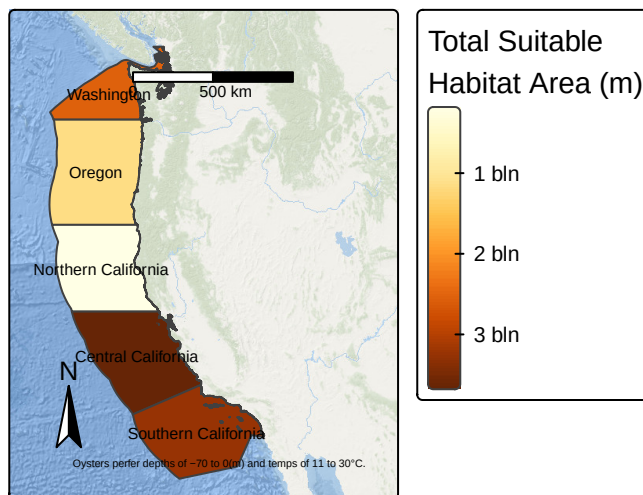
Suitable Depth & Temp for Oysters



Suitable vs Unsuitable Habitat



Suitable Habitat for Oysters within
the West Coast Exclusive Economic Zones



Suitable locations for Siliqua-patula

The Pacific razor clam location ranges from the Aleutian Islands in Alaska to Pismo Beach, California. The clam is typically discovered in depths between 0-55m, and its preferred temperature is around 5°C (estimating 0°C - 10°C).

Therefore, Pacific razor clam research has shown that the clams need the following conditions for optimal growth:

- Sea surface temperature: 0-10°C
- Depth: 0-55 meters below sea level

Use Function to Find Suitable Habitat for clams in West Coast EEZs

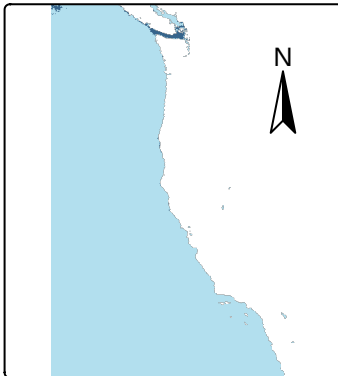
```
# Use function
df_suit_area_eez <- habitat(
  depth_df = bathy_updated,
  temp_df = sst_mean,
  depth_min = 0,
  depth_max = 55,
  temp_min = 0,
  temp_max = 10,
  species_name = 'Clams',
  eeze_df = eeze
)
```

```
[1] "CRS match"
```

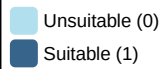
```
[1] "The area of all suitable Clams habitats in the West Coast EEZs is 0.24 x 10^9 m^2"
```

EEZ location	Suitable Habitat Area (m)
Oregon	6.34e+07
Northern California	1.66e+07
Central California	0.00e+00
Southern California	0.00e+00
Washington	1.58e+08

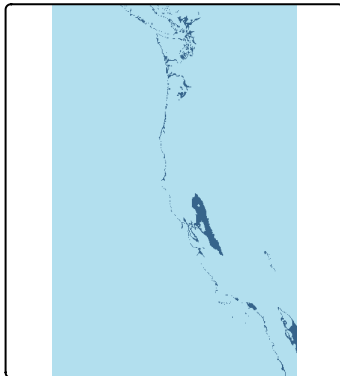
Suitable Temp for Clams



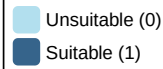
Suitable vs Unsuitable Temp



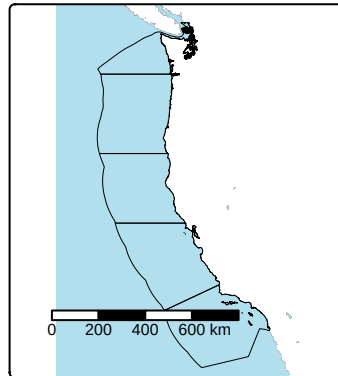
Suitable Depth for Clams



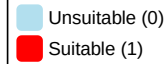
Suitable vs Unsuitable Depth



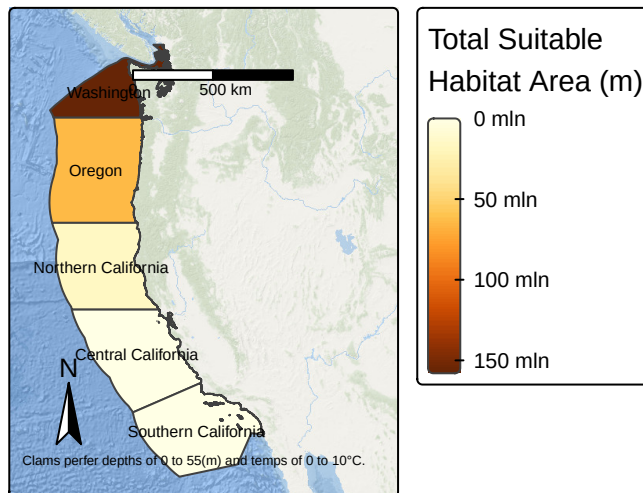
Suitable Depth & Temp for Clams



Suitable vs Unsuitable Habitat



Suitable Habitat for Clams within
the West Coast Exclusive Economic Zones



Additional Map For Clams

I wanted to see Washington better, so I created tailored map to look at Washington

```

# Creating dataframes in order to plot
rcl_depth_clam <- matrix(c(-Inf, -55, 0, # UNSUITABLE
                          -55, 0, 1, # SUITABLE HABITAT
                          0, Inf, 0), # UNSUITABL
                        ncol = 3, byrow = TRUE)

# Temp matrix
rcl_temp_clam <- matrix(c(-Inf, 0, 0, # UNSUITABLE
                          0, 10, 1, # SUITABLE HABITAT
                          10, Inf, 0), # UNSUITABL
                        ncol = 3, byrow = TRUE)

# Apply the reclassification matrix to the depth and temp rasters
depth_reclass_clam <- classify(bathy_updated, rcl_depth_clam)
temp_reclass_clam <- classify(sst_mean, rcl_temp_clam)

# Remove matrixs as objects
rm(list = c('rcl_depth_clam', 'rcl_temp_clam'))

# Multiple temp and depth rasters
clam_suit_raster <- depth_reclass_clam * temp_reclass_clam
# Crop to EEZs
clam_suit_crop <- mask(clam_suit_raster, eeZ)

# bbox of Washington EEZ
washington <- eeZ %>% filter(rgn_key == 'WA')
clam_suit_washington <- mask(clam_suit_crop, washington)

# Inset map of Just clams in Washington
washington_inset <- tm_shape(
  clam_suit_washington,
  bbox = sf::st_bbox(c(
    xmin = -129.5, xmax = -122,
    ymin = 45.5, ymax = 49
  )),
  crs = st_crs(clam_suit_washington) ) +
  tm_raster(
    col.scale = tm_scale_categorical(
      values = c('lightblue2', 'red')),

```

```

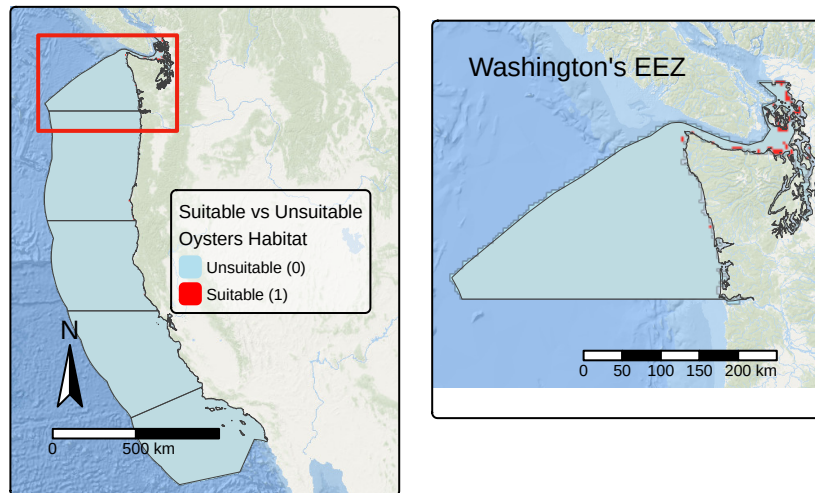
    col.legend = tm_legend_hide()) +
tm_shape(washington) +
tm_polygons(lwd = 0.35,
            fill = "peachpuff2",
            fill_alpha = 0.2) +
tm_title(text = "Washington's EEZ",
        position = tm_pos_in(),
        size = 0.8) +
tm_basemap("Esri.OceanBasemap") +
tm_scalebar(position = c('right', 'bottom'))

# TEMP AND DEPTH AND EEZ MAP
tm_shape(
    clam_suit_crop,
    bbox = sf::st_bbox(c(
        xmin = -131, xmax = -110,
        ymin = 30,  ymax = 50
    )),
    crs = st_crs(clam_suit_crop)) +
tm_raster(
    col.scale = tm_scale_categorical(
        labels = c("Unsuitable (0)", "Suitable (1)"),
        values = c('lightblue2', 'red'),
    ),
    col.legend = tm_legend(title = "Suitable vs Unsuitable\nOysters Habitat",
        position = tm_pos_in('right', 'center'),
        title.size = 0.6,
        text.size = 0.5,
        bg.alpha = 0.5)
    ) +
tm_title(text = "Suitable Depth & Temp for Clams\nwithin the West Coast EEZs") +
tm_shape(eez) +
tm_polygons(lwd = 0.35,
            fill = "peachpuff2",
            fill_alpha = 0.2) +
tm_basemap("Esri.OceanBasemap") +
tm_compass(position = c('left', 'bottom'))+
tm_scalebar(position = c('left', 'bottom')) +
tm_inset(washington_inset,
        position = tm_pos_out(),
        width = 15,

```

```
height = 15)
```

Suitable Depth & Temp for Clams within the West Coast EEZs



```
# Remove objects
rm(list = c('map_temp_clam', 'map_depth_clam', 'map_clam', 'clam_suit_washington', 'washingt
```

The Pacific razor clam requires cold temperatures ranging from 0 to 10°C and shallow depths from 0 to 55m. This pins the clams to the coastline and to the North. Hence, Washington's Exclusive Economic Zone has the most suitable habitat for the Pacific razor clam.

References

Hall, S. J., Delaporte, A., Phillips, M. J., Beveridge, M. & O'Keefe, M. Blue Frontiers: Managing the Environmental Costs of Aquaculture (The WorldFish Center, Penang, Malaysia, 2011).

Gentry, R. R., Froehlich, H. E., Grimm, D., Kareiva, P., Parke, M., Rust, M., Gaines, S. D., & Halpern, B. S. Mapping the global potential for marine aquaculture. *Nature Ecology & Evolution*, 1, 1317-1324 (2017).

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